

EIVINAS BUTKUS

PhD Student in Psychology at Columbia University, supervised by Niko Kriegeskorte.

Studying how resource constraints—limitations in space, time, energy and data—shape natural and artificial intelligence, and how brains and machines build internal models of the world.

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Education

- 2021– **Columbia University** · PhD Psychology (defense in May 2027)
Thesis: *Space, Time, Energy, and Data: How Resource Costs Shape Neural Computation*
Advisor: Niko Kriegeskorte
- 2018–2019 **Imperial College London** · MSc Computer Science (Distinction)
Thesis: *Dimensionality Reduction Techniques for Automatic Behavior Descriptors*
Advisor: Antoine Cully
- 2015–2018 **King's College London** · BA Philosophy (First Class Honours)
Thesis: *Models, Understanding, and Artificial Intelligence*
Advisors: Patrick Butlin and David Papineau
- 2014–2015 **Vilnius University** · BSc Mathematics (one year completed)

Employment

- 2019–2021 **Yale University** · Department of Psychology · Research Assistant
Co-developed a computational model of human attention in dynamic scenes.
Advisor: Ilker Yildirim

Experience

- 2026– **Cognitive Computational Neuroscience (CCN) 2026** · GAC Organizer
Leading a Generative Adversarial Collaboration (GAC) on “Do world models emerge in neural networks trained for prediction?” with 12 co-authors.
- 2024– **NSF AI Institute for Artificial and Natural Intelligence (ARNI)** · Trainee
Co-organizing the working group on multi-resource cost optimization in neural networks; leading a perspective article on the same topic with 13 co-authors.

- 2025–2026 **Cognitive Computational Neuroscience (CCN) 2026** · Chair, Local Committee
Led a six-person committee delivering social programming for ~1000 conference attendees, plus volunteer coordination and local resources for the conference website.
- 2025–2026 **The Columbia University Seminars** · Rapporteur
Took minutes for the Cognitive and Behavioral Neuroscience Seminar.
 Chairs: Christopher Baldassano and Herbert S. Terrace.

Publications

* *equal contribution*

Peer reviewed

- Adaptive computation as a new mechanism of dynamic human attention
 M. Belledonne, E. Butkus, B. Scholl, & I. Yildirim (2026) · *Psychological Review*
- Causal Discovery and Inference through Next-Token Prediction
 E. Butkus & N. Kriegeskorte (2025) · *Neural Information Processing Systems (NeurIPS)*

In revision or under review

- How attention saves energy in vision
 E. Butkus, Z. Ying, & N. Kriegeskorte (2026) · *bioRxiv* — *in revision at Nature Neuroscience*
- Growing a Neural Network in Breadth, Depth, and Time
 E. Butkus, K. Garzón Gupta, & N. Kriegeskorte (2026) · *arXiv* — *under review*

In preparation

- Space, time, energy, and data: A multi-resource cost framework for neural computation
 E. Butkus, J. Achterberg, N. M. Blauch, D. B. Chklovskii, T. L. Griffiths, T. C. Kietzmann, G. W. Lindsay, X. Pitkow, P. R. Schrater, A. A. Stocker, X.-X. Wei, Z. Ying, R. Zemel, & N. Kriegeskorte · *perspective article*
- Do world models emerge in neural networks trained for prediction?
 E. Butkus, A. Tartaglini, E. Su, S. LeGris, Z. Ying, S. Joseph, B. Cheung, P. Butlin, J. Hu, J. Morrison, I. Yildirim, B. Lake, & N. Kriegeskorte · *CCN Generative Adversarial Collaboration position paper*

Workshop papers

- Human-like multiple object tracking through occlusion via gaze-following
 B. Peters*, E. Butkus*, & N. Kriegeskorte (2024) · *Gaze Meets ML Workshop (NeurIPS)*
- How do humans and machine learning models track multiple objects through occlusion?
 B. Peters, E. Butkus, & N. Kriegeskorte (2022) · *SVRHM Workshop (NeurIPS)*

Conference abstracts

- Blame via Inverse Reinforcement Learning
E. Butkus*, C. Mott*, N. Kriegeskorte, & C. Baldassano (2026) · *Cognitive Computational Neuroscience (CCN)*
- Attention enables flexible energy-efficient vision
E. Butkus, Z. Ying, & N. Kriegeskorte (2026) · *Vision Sciences Society (VSS)*
- Recurrent Attentional Selection Can Explain Flexible Trading of Accuracy and Energy in Biological Vision
E. Butkus, Z. Ying, P. Chen, & N. Kriegeskorte (2024) · *Cognitive Computational Neuroscience (CCN)*
- FlyingObjects: Testing and aligning humans and machines in gamified object vision tasks
B. Peters, E. Butkus, M. H. Retchin, & N. Kriegeskorte (2024) · *Vision Sciences Society (VSS)*
- Adaptive recurrent visual inference with learnt top-down attention
E. Butkus & N. Kriegeskorte (2024) · *Computational and Systems Neuroscience (Cosyne)*
- Attentional dynamics during multiple object tracking are explained at subsecond resolution by a new ‘hypothesis-driven adaptive computation’ framework
M. Belledonne, E. Butkus, B. Scholl, & I. Yildirim (2021) · *Vision Sciences Society (VSS)*
- Modeling temporal attention in dynamic scenes: Hypothesis-driven resource allocation using adaptive computation explains both objective tracking performance and subjective effort judgments
E. Butkus*, M. Belledonne*, B. Scholl, & I. Yildirim (2020) · *Proceedings of the Annual Meeting of the Cognitive Science Society (CogSci)*

Talks

- Do world models emerge in neural networks trained for prediction?
Generative Adversarial Collaboration (GAC) at Cognitive Computational Neuroscience (CCN) 2026 — Introduction — joint with John Morrison · August 2026
- Causal Models and Causal Reasoning Can Emerge Through Next-Token Prediction
GAC on Emergent World Models at CCN 2026 · August 2026
- How Attention Saves Energy in Vision
Harvard Vision Lab, Harvard University · June 2026
- How Attention Saves Energy in Vision
Kietzmann Lab, University of Osnabrück (invited) · June 2026
- Blame via Inverse Reinforcement Learning
Six College Conference, Yale University · May 2026
- Attention Enables Flexible Energy-Efficient Vision
Michael Woodford Group, Columbia University · January 2026
- Attention Enables Flexible Energy-Efficient Vision
Symposium on Computational Modeling of Human Attention at CogSci 2025 (invited) · July 2025

- Causal Discovery and Inference Through Next-Token Prediction
Causal Artificial Intelligence Lab, Columbia University · May 2025
- Attention Makes Vision Energy-Efficient
Cognitive and Neural Computation Lab, Yale University · May 2025
- Keynote on NeuroAI
Meeting of Young Lithuanian Mathematicians · December 2024
- Modeling Dynamic Attention: Hypothesis-Driven Resource Allocation Using Adaptive Computation
42nd Annual Conference of the Cognitive Science Society · August 2020

Teaching

Lectures

- Lecture on Cognitive Computational Neuroscience
UN2430 Cognitive Neuroscience, Columbia University · May 2026
- Lecture on Cognitive Science and AI
Marius Jakulis Jason Foundation · June 2020

Course Teaching Assistance

- Machine Intelligence · Columbia University · Fall 2026
- Cognitive Neuroscience · Columbia University · Spring 2026
- Drugs & Behavior · Columbia University · Fall 2024
- Statistics for Behavioral Scientists · Columbia University · Fall 2023
- Research Methods: Human Behavior · Columbia University · Spring 2023
- Science of Psychology · Columbia University · Spring 2022

Tutorials

- Statistical inference on representational geometries · *Neuromatch NeuroAI Course* · July 2024
- Probabilistic programming using Gen · *Workshop at Yale Psychology* · December 2019

Service

- Journal reviewer: *Nature Communications*
- Conference reviewer: *NeurIPS* · *Cognitive Computational Neuroscience (CCN)* · *Cognitive Science Society (CogSci)* · *NeurIPS Mechanistic Interpretability Workshop*

Mentorship

- Manush M. Kalwari · Postbac Research Assistant in Kriegeskorte Lab · Columbia University · February–June 2026
- Peiyu Chen · Postbac Research Assistant in Kriegeskorte Lab · Columbia University · Spring 2024

Training

- Computational and Cognitive Neuroscience · Summer School · *Cold Spring Harbor Asia* · June 2024

Funding & Awards

- Cosyne Travel Grant · Columbia University · 2024
- Scholarship for MSc at Imperial College London · Lithuanian Foundation · 2018 · \$3,500
- Selected for Talents for Lithuania Fund (crowdfunded scholarship) · 2018 · €2,000

Skills

- Proficient in Python, Julia, C#. Deep learning (PyTorch), probabilistic programming (Gen.jl, Pyro), high-performance computing (Slurm).

Extracurricular

- Visual Computational Neuroscience Journal Club at the Zuckerman Institute · Co-organizer · New York, USA · 2021–2025
- Debate Mate · Teaching debating to kids from underprivileged schools · London, UK · 2018–2019
- Academic Buddy · Helping Lithuanian high school students apply to UK universities · Lithuania · 2018–2019
- Kripkenstein · Weekly reading group on Kripke's interpretation of Wittgenstein, King's College London · 2017

Press & Media

- Interview on my path from philosophy to artificial minds
Study International · January 2026
- Alumni profile on the path from philosophy to AI and cognitive science
King's College London · May 2020
- Interview about the Talents for Lithuania Fund crowdfunding campaign
Lithuanian National Radio and Television (LRT) · August 2018